

Name: TELLAPATI VENKATA KARTIK REG NO: 99230040904

1. If an API returns 429, what should your program do?

If an API returns 429, it means “Too Many Requests.” This happens when our program sends too many requests to the API in a short time.

The program should not send another request immediately. It should wait for some time and then try again. If the API still returns 429, the program should wait longer before trying again.

For example, the program can wait for 1 second, then 2 seconds, then 4 seconds. This helps avoid putting too much load on the API.

2. If response.json() fails, what should your program do?

response.json() is used to convert the API response into JSON data. Sometimes the response may not contain valid JSON. In that situation, response.json() can produce an error.

The program should handle this error using try-except. This prevents the program from crashing and allows us to display a suitable error message or handle the response in another way.

```
try:
    data = response.json()
except:
    print("Response is not valid JSON")
```

This makes the program safer because it can continue working even when the response is not in the expected format.

3. If a JSON key is missing, why might .get() be safer than []?

When we use data["name"], Python expects the "name" key to be present. If the key is missing, Python will give a KeyError and the program may stop.

Using data.get("name") is safer because, if the key is missing, it returns None instead of causing an error.

We can also give a default value:

```
name = data.get("name", "Unknown")
```

If the "name" key is missing, the program will use "Unknown".

.get() is safer when a JSON key may be missing because it helps prevent errors and allows the program to handle missing data properly.